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National Capacity Building

The biggest secret of the Israeli high tech system is the military's ability to look at people while they are in high school.

Nadav Zafrir, former Commander of Unit 8200, now high tech entrepreneur

Technological prowess played a critical role in Israel's national security doctrine and economic policy from the earliest days. The cyber realm, with its emphasis on outstanding scientific and technological creativity and innovation and potential for rapid and high returns on investments of a comparatively modest scale, was considered to be particularly suited to Israel's national strengths. It was also particularly suited to Israel's strategic culture and national temperament, or what we call chutzpah gone viral. Over the years, these basic national cultural characteristics fused with the strategic and economic imperatives to improvise and innovate. In the process, they became deeply ingrained, almost reflexive Israeli traits, a national modus operandi and sphere of excellence, often manifested even when more established and routinized modes of operation might be preferable. The IDF and intelligence agencies, which have an unusually symbiotic relationship with the civil cyber sector,[1] are also deeply imbued with this innovative national culture and have become primary engines thereof, further promoting cyber and Israeli high tech in general.

It is here that the constructivist concept of strategic culture meets the realist concept of creative insecurity. Strategic culture, as already noted, refers to the impact of a state's historical beliefs, collective memories, values, traditions, mentality, and strategic assumptions on its national security decision-making.[2] Creative insecurity arises when significant external threats, economic and/or military, incentivize scientific and technological innovation designed to foster and sustain an internationally competitive economy, thereby enabling the state to purchase the means to defend itself or build domestic defense industries.[3]

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Both concepts are highly applicable to Israel, which recognized from the beginning that the cyber realm's unique blend of civil, commercial, criminal, counterterrorism, and military dimensions required governmental leadership and a "whole of society"[4] response. To this end, a variety of policies and programs have been developed to foster Israel's overall cyber capabilities—in sum, a national cyber ecosystem. Developing this ecosystem has been a basic aim of Israel's cyber policy and of every cabinet decision in this regard ever since the first one was adopted in 2002.[5]

Chapter 8 has three sections. The first presents an overview of Israel's civil cyber ecosystem, tracing such issues as investment in cyber R&D, comparative indices of Israeli innovation, the size of Israel's high tech and cyber sectors, numbers of cyber firms, and more. The second section addresses some of the reasons for Israel's success in the high tech and cyber areas and is divided into four subsections: cyber R&D and technology spillovers, development of cyber human resources, Israel's innovative cyber culture, and social networks. The chapter concludes with a discussion of some of the clouds on Israel's high tech and cyber horizons, possible misdeeds in the cyber realm, and new opportunities for technological development.

Building Cyber Industry: From Jaffa Oranges to Silicon Wadi

The reason why Israel has come to be known as the startup nation clearly emerges from the numbers. Israel's economy is the most tech-dependent in the world, with 13% of GDP and 31% of exports originating from the high tech sector. Israel has long been ranked first in the world in terms of investment in R&D and venture capital, as a percentage of GDP, approximately double the OECD average. If one adds military R&D, the gap is even larger, with an additional 1–1.5% of GDP.[6]

Israel has the largest number of high tech startups per capita in the world,[7] with approximately 600 new ones established on average each year, making a total of roughly 6,000 in 2019. Israel has the second largest number of firms listed on the NASDAQ of any country outside of North America.[8] In 2020 Israeli startups raised $11.5 billion, 20% more than 2019 and more than four times the amount raised a decade earlier.[9] In the past, Israeli high tech firms were considered to be excessively focused on short-term gains and a rush to go public, rather than taking a longer-term perspective designed to build a significant international presence. This is no longer the case. Israel already had more "unicorns" (firms with a worth over $1 billion) per capita in 2020 than any other country in the world, indeed, almost double the rate in the United States; and with just 0.1% of the global population, it had one third of all cyber security unicorns. While only 10% of startups globally usually survive more than five years, in Israel the comparable figure is a whopping 65%.[10]

In 2020 the World Economic Forum ranked Israel the sixth most innovative nation in the world, down from second place in 2016–2017, but still very high. The study was based on 12 indices of competitiveness, including innovation, technological readiness, business sophistication, higher education, patent filing, R&D expenditure, and more. In 2021 a Bloomberg study, based on seven indices of innovation, placed Israel in seventh place, down two notches from where it had been two years earlier, but still ahead of its tenth-place ranking during the two years prior to that.[11]

Israel has one of the world's highest concentrations, per capita, of technologically advanced human resources. It has the highest concentration of scientists of any country, 135 for every 10,000 people, compared to 85 in the US; ranks first in the world per capita in research personnel and third in university graduates; and is consistently ranked among the top ten in patent applications.[12]

There are nearly 400 multinational R&D centers in Israel, including more than 25 in the cyber realm alone. On average, 22 additional R&D centers open each year.[13] The list of multinational corporations that have set up R&D centers in Israel reads like a Who's Who of the top firms in the field.[*] These firms employ tens of thousands of people and are involved in the development of highly advanced systems. Microsoft's Israeli R&D center, for example, is one of three "strategic global development centers" and is considered home to some of the company's most innovative technologies, with a focus today on big data, business intelligence, cloud storage, and artificial intelligence. Many of these multinational companies have also bought numerous Israeli startups.

In recent years Israel has become a growing center of startups in the "smart transportation" sector. Global giants, such as General Motors, Ford, Renault-Nissan-Mitsubishi, Hyundai, and Volkswagen, among others, have established research centers in Israel dealing with autonomous mobility, vehicle technology, and more.[14] Israel is also poised to become a significant player in what has been dubbed the Fourth Industrial Revolution, also known as Industry 4.0, the Industrial Internet of Things (IIoT), and smart manufacturing. As of 2019, 230 companies were active in Industry 4.0 in Israel, an increase of 60% over 2014, including 23 R&D centers, 11 hubs, and 8 accelerators and incubators. Venture capitalists have taken note of the potential, and venture capital financing for

[*] Intel, Google, Microsoft, IBM, Amazon, Facebook, Deutsche Telekom, HP, Cisco, Marvell, Apple, McAfee, EMC, PayPal, Oracle, General Electric, and Lockheed Martin, among others.

these firms in 2018 accounted for 5% of all global financing in the field. The government has also earmarked over $100 million to support the transition of the local manufacturing industry to IIoT.[15]

Israel's high tech sector employs approximately 300,000 people, or 9% of the total national labor force, including 50,000 software engineers. The cyber industry employs 20,000 people (not including defense firms), of whom 7,000–8,000 are engineers.[16] The high tech sector is based around four primary hubs—Tel Aviv, Haifa, Jerusalem, and Beersheba—and a number of smaller satellite ones, but really all constitute one region. The greater Tel Aviv hub, the heart of the Israeli high tech ecosystem, is widely considered the second most important in the world, after Silicon Valley. Much of the activity is centered on Tel Aviv and Bar Ilan Universities, the Weitzman Institute, and the many advanced IDF bases in the area. More surprisingly, Jerusalem, with its large traditional and ultra-Orthodox populations, has become a flourishing center for biomed, cleantech, Internet/mobile startups, accelerators, and the support of service providers. The total number of high tech firms in Jerusalem doubled between 2012 and 2020, many of them centered around Hebrew University.[17] Haifa, which blazed the way in the early years of Israel's high tech revolution, has lost its leading role, although the Technion remains Israel's premier institute of higher education in technological fields.

As early as 2015 Israel had over 300 startups focused on the cyber realm, double the number five years earlier and, remarkably, equal to the total number of such firms in the rest of the world combined, not including the United States. By 2019 the number of cyber startups had grown to 436, out of all 752 cyber firms in Israel, making it the world's second largest exporter of cyber security software.[18]

Between 2013 and 2017 Israeli firms accounted for 7% of all global cyber security trade, ahead of Britain, Canada, and China.[19] In 2016 Israeli cyber exports reached $6.5 billion, equal to 8–10% of a global market estimated to be worth roughly $70 billion;[20] in 2021 they were worth $11 billion.[21] In terms of foreign direct investment (FDI) in cyber security firms, Israel is again second only to the United States. Between 2013 and 2016, FDI in Israeli cyber security firms grew from $165 million, about 11% of the world total, to $500 million, or 15% of the world total at the time.[22] In 2020, 31% of cyber investment worldwide was in Israel, an increase of more than 70% over the previous year and a 50-fold increase over the previous decade. Altogether, over 20 cyber firms were purchased with a combined value of some $4.7 billion. Israel never previously enjoyed such dominance in any technological area.[23]

The hub in Beersheba starts at a disadvantage, given its distance from the center of the country and less attractive desert environment, but the government has officially declared its intention to turn the city into Israel's "cyber capital." Beersheba's Advanced Technology Park (ATP), called Cyber Spark, was established in 2013 as a joint venture of the government, local municipality, and Ben-Gurion University (BGU) and is at the heart of these plans. It is also an important example of the national policy of promoting heightened cooperation in the cyber realm between academia, multinational corporations, startups, government agencies, and the IDF, all of which work in the ATP office complex and have the opportunity to assist each other with knowledge, personnel, and resources and to foster new innovative ideas. To encourage firms to open offices in the ATP, the government offers grants covering up to 20% of salaries. In 2019 some 70 firms had offices there, ranging from global giants such as Lockheed Martin, IBM, Deutsche TeleKom, and EMC to small startups. The ATP also houses the INCD's national cyber emergency response team (CERT-IL) and personnel from a number of IDF units and the ISA.[24]

The IDF is slated to play an important role in Beersheba's cyber ecosystem. Its new national cyber center, adjacent to the ATP, is scheduled to be completed by 2023. The center will house the C4I and Cyber Defense Branch's[†] technological units and computer, cyber and communications schools, as well as some Air Force and other units. The thousands of soldiers serving at the center will be able to register for courses at BGU.[25] The IDF also plans to relocate intelligence units, including Unit 8200 (Israel's equivalent of the US National Security Agency), with a combined staff of approximately 19,000 people, to a new base near Beersheba.[26]

Israel's unique cyber ecosystem is based on unusually close collaboration between the government, defense establishment, academia, and commercial sectors.[27] In an area changing as rapidly as the cyber realm, in which a technological generation is no more than 1.5 years, it is impossible to predict all threats and opportunities. Israel's cyber ecosystem was thus designed to be highly flexible and capable of evolving in accordance with the frenetic pace of technological change.[28] To this end, while the government was deeply involved in promoting the cyber ecosystem in its early years, especially industry and academia, it has become essentially autonomous in the interim, capable of standing on its own. Indeed, the INCD has concluded that the cyber industry no longer needs significant governmental funding, given the large sums readily available from the private market, and it is now exploring new ways in which the government can best be of help.[29]

[†] C4I—command and control, computers and communications. The branch is responsible for IDF communications and computer systems and cyber defense.

Cyber R&D and Technology Spillovers

From the beginning, cyber R&D (academic, commercial, and military) has been one of the primary means by which Israel has sought to achieve and maintain a lead in the global cyber realm.[30] In the academic area, the INCD matched the universities for a total investment of some 250 million shekels in order to encourage the establishment of cyber centers at each one. Each university was free to decide where its strengths lay and which areas it wished to focus on.[‡31]

Ever since 2012, the INCD has provided hundreds of millions of shekels to Israel's universities for research. In 2013, there were approximately 30 academic cyber researchers in Israel; by 2019 the number had reached 200.[32] The INCD and Israel Innovation Authority (IIA)[§] fund programs designed to promote cyber R&D and entrepreneurship. Of 2,875 R&D funding requests submitted to the IIA in 2018, it invested in 920 firms and provided 1.7 billion shekels (nearly $500 million) in funding for approximately 1,500 projects.[33] The IIA further provides direct assistance to firms engaged in cyber R&D, as well as matching grants with third parties, both Israeli and foreign. Unlike INCD projects, which typically focus on governmental or military cyber capabilities, IIA assistance is designed solely for commercial projects and dozens have been funded with firms from states around the world.[**] The IIA and Ministry of Science and Technology also offer scholarships and research grants in the area of cyber security.[34]

Israel further seeks to promote cyber R&D through cooperation with leading multinational high tech firms. Deutsche Telekom, for example, which is already involved in the Cyber Spark ATP in Beersheba, has expanded its collaborative efforts with BGU, with a focus on network security and big data. IBM established a research center there focused on big data, cloud computing, cognitive cyber protection, security of connected vehicles, and biometric authentication. Fujitsu established a cyber security center at BGU focusing on developing security technologies for AI-based systems. Germany's Fraunhofer Institute for

[‡] The cyber center at BGU, Cyber@BGU, focuses on technology and applied science. Hebrew University's Cyber Security Research Center specializes in security and cryptography, as well as international law in the cyber field. The Technion's Hiroshi Fujiwara Cyber Security Research Center focuses on software and hardware protection, as do Haifa University's Center for Cyber Law and Policy and Bar-Ilan University's Research Center in Applied Cryptography and Cyber Security. Tel Aviv University's Blavatnik Interdisciplinary Cyber Research Center takes a broader interdisciplinary approach, including policy and legal issues.

[§] Formerly known as the Office of the Chief Scientist, IIA is under the Ministry of the Economy and Industry.

[**] Among others, the United States, Singapore, South Korea, Canada, China, Japan, India, Australia, and Latin America, in addition to a number of multilateral European projects.

Secure Information Technology established a joint cyber research center with the Hebrew University in Jerusalem.[35]

The IDF and Cyber R&D—the contribution of the IDF, intelligence agencies and the defense establishment, as a whole, to Israel's high tech capabilities, cannot be overstated. Much like the Department of Defense in the US, and especially the unique role played by DARPA (the Defense Advanced Research Projects Agency), the defense establishment in Israel and its DARPA equivalent, MAFAT (the Directorate of Defense R&D), have been among the driving forces behind innovation in Israel.[36]

In practice, the MoD, IDF, especially Military Intelligence Units 8200 and 81 (the advanced technologies unit), and various IAF and C4I Branch units have become important incubators and accelerators of high tech startups, with knowledge, capital, and resources flowing to the private sector—and vice versa. This is especially true in the cyber realm, where the defense establishment's deep involvement in defense related cyber R&D has spawned numerous civilian applications. Many of the leading high tech and cyber firms in Israel—Check Point, Palo Alto Networks, and NICE Systems to mention just a few—were founded by people who had served in Units 81 and 8200 and other leading IDF units.[37]

Intelligence Unit 81 is the most decorated unit in the IDF, having won an extraordinary 36 prestigious annual Israel Defense Prizes for technological innovation. A top-secret unit whose motto is "turning the impossible into the possible," Unit 81 is at the forefront of global technology. Approximately 100 veterans of the unit, who served between 2003 and 2010, founded 50 startups during the following decade alone, with accumulated valuations exceeding $10 billion. One unit veteran, who founded a number of highly successful startups, explains this unusual entrepreneurial success rate this way: "A team that has worked together before is a substantial force multiplier. You can't find that in Silicon Valley, either in terms of the pool you can draw from, or in terms of knowledge and experience. Even Google and Microsoft don't have a concentration of talent like Unit 81."[38]

The IDF, IAF, and MoD, in collaboration with a private firm, launched an innovation center in 2019—dubbed INNOFENSE—designed to promote the development of new technologies and innovative startups. The center focuses on civilian cyber technologies with military applications, such as big data, the IoT, unmanned systems, robotics, cyber, deep learning, homeland security, and border security.[39] The IDF also launched two other programs, Stargate and Star Trek, the first of ten planned programs that apply AI to interpret intelligence information for operational purposes. In the past, sensitive and sophisticated software of this sort would have been developed by the IDF solely in-house, but the need for personnel with unique expertise and the push for more rapid development times has led to growing cooperation with commercial firms. The ability to use the sophisticated code developed—minus the sensitive defense applications—is the private firms' primary commercial incentive for participating in these programs.[40]

The Mossad, similarly, has established Libertad, a technology innovation fund that invests in startups in a variety of areas, including financial tech, robotics, AI, drones, remote personality analysis, natural language processing, voice analysis and processing, synthetic biology, Blockchain, and online privacy. Investments are made on particularly attractive terms for the entrepreneurs; in return, the Mossad gains access to unique intellectual property, without retaining commercial rights after the R&D stage is over.[41]

Cyber Human Resource Development

In the early 2010s the INCD,[††] IDF, Ministry of Education, and others concluded that a severe shortage of highly trained technological personnel had become a primary obstacle to Israel's future growth in the cyber realm and high tech generally, as well as to its rapidly expanding needs in the defense area. The resulting INCD strategy for the development of human resources in the cyber field, designed to expand the overall national pool of technological personnel, was based on a three-tiered pyramid of skills: at the bottom a broad-base of information and communications technology personnel generally, a second layer of more highly trained cyber specialists, including those in the defense area, and on top a smaller number of truly exceptional R&D experts, those who account for most of the important breakthroughs.[42]

To increase the national pool of technological personnel, three further decisions were made. The first was to begin educational programs in schools at as early an age as possible.[43] Research demonstrated that students who had studied computers in high school were far more likely to continue working in the field in the future, even if they did not serve in computer-related units in the IDF. Among those who did serve in such units, the numbers were even higher.[44]

The second decision was to make a special effort to reach out to population groups that were underrepresented in technological fields and which were likely to remain so in the absence of remedial efforts. A primary target audience in this regard was young people from rural and disadvantaged areas, known in Israel as the periphery, as well as women, ultra-Orthodox Jews, and Israeli Arabs.[45] To this day, despite the extensive programs outlined later, the high tech sector is still

[††] In its earlier incarnation as the INCB.

comprised of approximately two-thirds Jewish men, one-third Jewish women; only about 3% are ultra-Orthodox and 2% Arabs.[46]

The third decision was to enlist Israel's high tech sector and academic institutions in the effort to increase the overall national pool of cyber personnel. IDF training programs, impressive though they already were at the time, were insufficient to meet its own needs let alone to propel the entire ecosystem forward. Industry was found to be a particular bottleneck. Most cyber positions require three or more years of experience, but the costs of training new personnel on the job are very high. As a result, Israeli industry adds too few new people each year, despite the overall shortage in personnel.[47]

The INCD's well thought out human resources development strategy notwithstanding, the various cyber educational programs described later had their origins in a typically ad-hoc and informal but effective Israeli fashion. A few IDF officers and cyber executives, who knew each other, got together, identified the need, and began educational programs in a number of schools that were interested in participating. Unsurprisingly, the early programs focused primarily on the IDF's needs, especially mathematics, English, and programming.

Public School and Adult Education Programs—in 2016 the Ministry of Defence and INCD assumed formal responsibility for the programs and a national Cyber Education Center (CEC) was established. In addition to increasing the size of Israel's overall pool of cyber personnel, the CEC was charged with promoting social change through cyber education. Unlike the Ministry of Education, which is structured to address the needs of the school-age public as a whole, both in terms of the quality of teachers and level of instruction, the CEC was to hire advanced scientific and technological personnel in order to provide after-school programs and schools (even universities) with up-to-date pedagogical guidance and address the needs of exceptional young people.[48]

The CEC now oversees a range of cyber education programs, from the sixth grade through high school, all of which prepare students for future positions in high tech firms and academia, as well the IDF and intelligence agencies.[49] One such program, Shift, provides seventh graders with their first exposure to a variety of cyber related technologies, such as coding, algorithms, computer graphics, app programming, information security, and artificial intelligence. Outstanding middle school students (grades 7–9) can continue with a more in-depth after school program, StarTech, which includes training in programming, mobile apps, graphic design, computer games, and more, or with OnTop, a two-year problem solving and coding program.[50] The three-year Gvahim (Heights) program for grades 10–12 provides students with the opportunity to obtain a matriculation degree in cyber studies, computer science, or mathematics. At the time of the program's founding, Israel was the only country in the world to offer high school students the opportunity to take matriculation exams in cyber studies.[51]

The highly competitive Magshimim (Dream Fulfillers) program provides university-level instruction in cyber studies to underprivileged high school student who have exceptional coding and hacking skills. Much like Gvahim, it initially started as an initiative of the IDF and a private foundation, but rapidly became a national program and is now offered in 25 different locales, with 142 active classes. Like Gvahim, Magshimim is for grades 10–12, but is an after-school program. Classes meet twice a week for three hours and students further commit both to ten hours of homework each week and to participate each year in two cyber workshops and a summer cyber camp. In the late 2010s, approximately 700 out of 3,500 applicants passed the stringent entrance exams each year, of whom over 450 typically completed the program. 2,800 students will have graduated the Magshimim program by the end of 2022. The IDF and MoD share in the program's budget,[52] a clear indication of the importance they attach to it.

Participants in Magshimim have so outperformed their peers that the IDF asked that it be expanded to middle school students, as well. A further indication of the program's success is that more than 30% of Unit 8200's cyber personnel are Magshimim graduates today, in contrast with the even more highly disproportionate share of soldiers from socioeconomically well-off homes in the past. Some 65% of Magshimim graduates serve in intelligence and technological units, including cyber, and are avidly sought out by private sector firms upon completion of their military service.[53]

Another program, Gesharim ("Bridges"), is specifically designed for 7th to 9th graders from underprivileged communities. Run jointly by Unit 8200 and the IDF Educational Corps, the program provides training in coding, website and app development, cyber security and AI. In so doing, it also seeks to strengthen students problem-solving capabilities and sense of self-confidence and worth. User-friendly and experiential training programs, developed by Unit 8200, are used. The pilot program, which just began in 2022, has proven highly successful, with the heads of local authorities and parents pressing to have the program expanded to their communities. In 2023, it is expected to reach 18,000 students. [54]

Odyssey is a four-year program for exceptionally gifted students who wish to pursue an academic degree in computer science, or cyber studies, while still in high school. Many graduates subsequently enter the IDF's program for gifted soldiers, Talpiot (see below), and go on to become academic researchers and CTOs. Although Odyssey is small in numbers, just a few tens of students each year in each of the participating universities, it has an outsized impact.[55]

The CEC has put in place a number of projects designed to reach female, ultra-Orthodox, and disabled students. The CyberGirlz program provides a virtual community and mentoring program to attract young women interested in computer and cyber sciences. The mentors—all female—include soldiers in IDF technological units, college students studying computer science and software engineers and programmers from high tech companies. An earlier and now discontinued program, Mehamemet (Gorgeous), was also designed to encourage young women to pursue technological studies, providing 2,500 participants a year with their first chance to create an app.[56] Mamriot (Taking Off) trains Orthodox women for cyber and technological positions in the IDF, or as part of National Service (a non-military alternative to the IDF).[57]

The Ministry of Education, CEC, and some of the leading multinational R&D centers in Israel, cosponsor an annual "Coding Olympics" (dubbed "Skillz"), designed to encourage students to study coding and learn more about the cyber realm.[58] The Israeli branches of Microsoft, Google, and other leading tech firms conduct hackathons in which teams compete over the best means of preventing and mitigating simulated attacks.[59] Unit 8200 holds competitions in which students are challenged to disrupt an "adversary's" server, thereby enabling it to assess their performance and recruit the best among them.[60] Competitions such as these have been found to be an effective means of providing large numbers of young people with their first exposure to the world of programming. On-line courses for school age kids are another highly popular means of exposing them to programming and cyber generally.[61]

Between 2010 and 2023 the CEC will have spent approximately 250 million shekels[‡‡] on these programs, mostly for Magshimim. Of this, 90 million shekels will have come from a number of philanthropic foundations, the rest from the IDF, MoD, INCD, and other agencies.[62] Certainly by Israeli budgetary standards, these are impressively large sums.

Cyber security courses are now offered at every university in Israel, in addition to computer science and computer engineering.[63] A variety of non-academic adult cyber education programs are also available. She Codes seeks to increase the percentage of female programmers in the high-tech sector from the current 20% to 50%. Originally the initiative of a not-for-profit organization, the program soon gained government funding and now provides free training in coding for women in 40 centers around Israel. 4,000 women participated in the program in 2019 alone.[64] The Adva (Ripple) program, an unlikely joint endeavor of the Jerusalem municipality, tech giants such as IBM, Google, and Western Digital, religious seminaries, and philanthropic organizations, helps

‡‡ In recent years the shekel has fluctuated between 3.1 to 3.5 to the dollar.

impoverished women from the ultra-orthodox community obtain undergraduate degrees in computer science and mathematics. Although still small, only 60–80 graduates in each of the program's first two years, it is considered a success and projected to grow rapidly.[65]

Rapid change is also underway among Israel's Arab population. Whereas there were a mere 350 Arab engineers in the high tech sector in 2008, of whom only a handful were women, by 2019 the number had skyrocketed to 6,600, of whom 25% were women. Between 1984 and 2014 only 50 Arab students, on average, obtained undergraduate degrees in high tech disciplines each year. In the 2018–2019 school year alone, 4,553 Arab students began academic studies in these areas. In 2022 the IIA began a $70 million five-year program to promote high-tech in the Arab sector[66].

Another program, Israel Tech Challenge, seeks to leverage Israel's image as a world leader in cyber and high tech to promote immigration of diaspora Jews. The program, which was funded by the INCD and run by the Jewish Agency, a semi-governmental body, connects applicants who have degrees from leading foreign universities, with potential employers in Israel, even before they arrive in Israel. To date, the program has succeeded in attracting hundreds of people specializing in data science and cyber security, mostly from the US and France.[67]

The IDF and Cyber Human Resources Development—compulsory military service is a primary source of Israel's high tech prowess and at the heart of the unique Israeli nexus of the IDF, academia, and industry. Compulsory military service enables the IDF to harness the talents of Israel's best and brightest, essentially for free, for a number of years and is viewed as nothing less than critical to Israel's cyber success, without which the IDF would not have access to most of them, who would be drawn into the private sector.[68] It also means that the total talent pool available to Unit 8200, for example, is unusually large, especially given Israel's otherwise diminutive size, 1,000–2,000 of Israel's very best each year.[69] Whereas the United States' NSA has approximately 40,000 personnel, Unit 8200 reportedly has as many as 10,000. This is significantly smaller than the NSA, of course, but not by orders of magnitude.[70] The deputy head of 8200 Uri Stav says that "the greatest present that we have is the high-quality personnel who get here every year. It is there inexperience and rapid turnover that contribute to rapid changes and rapid responses to changes in the environment. They grow up with a technology that is changing all time. They are less rigid and in many cases change come from below. . . Most of the unit changes professions every decade. . . We are in the process of transformation. . . We count on the fact that the veterans of computer, math and robotics olympics reach here, or related units, every year.[71]

A disproportionate number of these veterans then go on to found and/or run Israel's cyber security firms or become leading academic cyber experts.

Moreover, just a few geniuses play an outsized role and make all the difference in the cyber realm. Many of the IDF's top talents were already employed by high tech firms prior to their induction (at age 18), their skills are in great demand both by the multinational tech giants and Israeli firms and many would not have served had they not been required to do so.[72] The advantages of compulsory conscription come full circle at the end of soldiers' service, as well. Somewhere between a few hundred and approximately 1,000 top-notch cyber experts, the very best Israel has, are discharged from the IDF each year and join the ranks of Israeli industry and academia. These numbers are large even by international standards.[73] China's National Cyber Security School, for example, is scheduled to graduate a class of 1,300 students in 2022 and only plans on approximately doubling that at some indefinite point in the future.[74]

To find the very best and brightest, the IDF scours Israel's high schools long before graduates begin their military service and conducts intensive screening processes for the different units.[75] In practice, the IDF has a surplus of recruits who wish to serve in the cyber units today, including many who already have university degrees or work experience in leading tech firms. The IDF is thus able to focus on the top candidates in each age cohort and ultimately select the very best. The competition for these recruits among IDF units is fierce, with first pick going to Military Intelligence.[76] A former head of Unit 8200, now a high tech entrepreneur, puts it this way: "The biggest secret of the Israeli high tech system is the military's ability to look at people while they are (still) in high school." When combined with the hands-on experience they gain through military service, as well as Israel's cyber capabilities in academia and its high tech firms, the result, he believes, "sparks magic."[77]

One of the IDF's Talpiot program seeks out the top 2% of high school students each year. Only 10% of them pass the initial battery of tests, mostly in physics and mathematics, but even this select group is further winnowed down through two days of grueling personality and aptitude testing. Those accepted commit to serving in the IDF for at least nine years, during which time they pursue undergraduate and graduate degrees, undergo specialized military training, and are typically involved in major R&D projects, in such areas such as machine learning, data mining, programming, operating systems, communications networks, and information security. Approximately one-third of Talpiot participants choose to stay on in the IDF and pursue military careers, another third become academics, and many of the rest have gone on to become Israel's most successful high tech entrepreneurs.[78]

A new offshoot of Talpiot, Odem, was jointly launched in 2022 by the IDF, Mossad, and ISA, in conjunction with the Ministries of Defense and Education. The 12-year program, which begins in 10th grade with three years at a sleep-away high school, is followed by an undergraduate degree in electrical engineering at the Technion and six years of service in IDF, Mossad, or ISA technological units. The program attracts participants by offering a free and personally tailored course of studies, with a large support staff, and assures them of "enormous influence" during their military service. Autonomous systems are a particular focus of the program.[79]

Unit 81's selection process is as grueling as Unit 8200's. Some 10,000 people meet the initial criteria each year and several thousand reach the screening process, but only a few hundred end up serving in Unit 81 or similar units.[80] The Academic Reserves (akin to the US ROTC) is still another source of exceptional personnel. Approximately 1% of high school graduates are given the opportunity each year to pursue undergraduate degrees in computer science, mathematics, engineering, and other areas of importance to the IDF, prior to their compulsory service. In exchange, they are required to serve for an extended period, typically five years.[81]

The IDF's critical need for highly trained and innovative personnel has made it one of the driving forces behind cyber education in Israel.[82] In addition to support for educational programs in the public school system, a variety of IDF units provide training programs that cover most areas of computer sciences, including mathematics, programming, infrastructure and network administration, software testing, and cyber defense.[83] Altogether, about 10,000 soldiers a year participate in these programs,[84] an indication of their subsequent impact on Israel's high tech sector generally.

Ever since 2012 the IDF has held a "cyber defenders" course, in which soldiers are taught how to analyze military computers and networks in order to detect and prevent attacks.[85] The IDF has also developed cyber simulators to train soldiers how to protect critical assets and networks.[86] In one course, soldiers train on a model Sim City, complete with residential and commercial neighborhoods, a railroad system, airport, electric grid, nuclear reactor, stock market, military base, and missile defense system. The simulated scenarios include cyber attacks that disrupt the cooling system at the nuclear reactor; remotely take over trains and traffic lights; shut-down the city's electric grid; disable the radar system at the airport; take control of stock exchange computers and cause a financial crisis; and hack the missile defense system to launch a direct missile at the city. Soldiers are trained to react quickly to prevent the attacks and are confronted with the consequences when they fail to do so.[87]

Approximately 500–600 soldiers are trained each year in offensive cyber operations at the Ashalim school in Beersheba. Courses typically last 20 weeks, starting with programming languages, before moving on to the specific skill sets the soldiers will need in their future units. Many have been glued to their keyboards since childhood, some already have undergraduate degrees when drafted and others have worked in high tech firms. High school or university level courses in mathematics and computer studies are a prerequisite for admission in most cases.[88]

Of the trainees at Ashalim, 25–30% are graduates of Magshimim and on average 12.5% are women, a number that has grown over time. In an effort to encourage women, who have been found to perform better in less competitive learning styles, trainees at Ashalim are not allowed to compare their achievements with each other, but to their own past performance. In order to increase their capacity for self-criticism, trainees are encouraged to discover and correct their own mistakes on tests. They are also required to sign on for extra time in the IDF, beyond their compulsory service.[89]

The above programs have clearly had a positive, if still insufficient impact on drawing women into IDF cyber programs. Unit 8200, for example, has seen an increase from 5% to 25% in female officers at the level of captain to lieutenant colonel. The deputy head of the units is that "this is still far from the female potential. It is a problem at the national level and despite attempts to change it, the problem has not been resolved."[90]

IDF cyber training programs span the entire length of a soldier's compulsory service—and beyond. A highly select group participate each year in intensive pre-induction cyber courses, many of whom are then chosen to serve in the IDF's leading intelligence and technological units.[91] Prior to discharge, combat soldiers are offered the opportunity to participate in seven-week cyber immersion courses, designed to provide them with the skills needed to gain employment in the private cyber market.[92] A similar commercial course would last months and be very expensive, in many cases prohibitively so.

Another program, Maagalim ("Cycles"), provides pre-discharge combat soldiers with an even more intensive training course in Unit 8200. At the end of the program, half of the graduates then serve either in Unit 8200 or the C4I Branch, as regular (paid) personnel, for a period of up to two years. Some remain in the IDF, most are then discharged with highly attractive skills to the commercial market.[93]

One particularly innovative course, run by Intelligence Unit 9900, which is responsible for geospatial intelligence, including satellite and high-altitude surveillance images,[94] trains autistic soldiers to do the highly exacting work of image interpretation, a skill at which they have proven particularly adept. The program's success later led to the establishment of a cyber security training course for autistic soldiers, as well. A number of private firms support the program, providing professional mentors to train the soldiers and offering internships to graduates, potentially leading to long term employment.[95]

Another way in which the IDF contributes to Israel's high tech prowess is through the unusual professional experience and command responsibility that it provides soldiers at a very young age. At a time when many of their peers abroad are still enjoying the final years of youth in college, these young soldiers are gaining real-world leadership experience commanding technological or combat units and/or learning advanced professional skills. Not everyone can serve in positions such as these, of course, but over half of the entire national population and nearly the entire secular Jewish population do serve in the IDF. At the very least, they are exposed to advanced military systems and to the IDF's innovative culture. It is an intense, transformative, and maturing experience, and the result is a workforce with an unusual number of highly trained, motivated, and disciplined young people with a can-do approach to problem-solving.[96]

In order to retain high quality cyber personnel, the IDF has been forced to be creative and devise special models of military service for them. One such model, known as "industrial capsules," enables cyber personnel to work in private firms for given periods of time and then return to military service. Another model allows IDF cyber personnel to work part time, while going to university, or working for private firms.[97] In addition to improved retainment rates—the primary objective—these models also provide the IDF with the benefit of the experience gained by personnel in the private sector, while the latter, which suffers from a constant shortage of highly trained cyber personnel, benefits as well.

An Innovative Cyber Culture: Chutzpah Gone Viral

Israeli society has a number of cultural attributes that make it particularly suited to high tech generally and to the cyber realm in particular. Start with a deeply ingrained national propensity to challenge authority and reject accepted norms, practices, and wisdom. Add a refusal to take no for an answer and constant search for alternative means of achieving an objective, even if it requires surmounting obstacles normally thought to be impassable. Further, add a belief that dedication, hard work, and out-of-the-box thinking can enable one to achieve almost any objective, along with a high tolerance for risk taking.[98] We call it chutzpah gone viral.

Drawing on Jewish traditions that emphasize knowledge and critical learning, including yeshiva education, in which students have long been encouraged to dispute interpretations of the Torah and other sacred texts, these national traits are also an outgrowth of Jewish history. Throughout the millennia, a sense of discrimination and persecution at the hands of hostile Gentile authorities imbued Jewish tradition with a view of authority, and of rules generally, as obstacles to be overcome, or circumvented, not constructive measures to be obeyed for the common good.[99]

Further contributing to the propensity to challenge authority, Israel is a highly heterogeneous immigrant society, with a population stemming from more than 70 different nationalities and cultural backgrounds. Israel may be predominantly Jewish (although approximately 20% of the population is not), but Jews of Iraqi, Polish, Ethiopian, Russian, or American heritage, for example, share little in the way of a common social and cultural background.[100] Each group has its own unique experiences, values, and ways of doing things, resulting in a constant state of social and cultural tension, often conflict, and a refusal to accept the norms and rules preferred by others. One of the results of this ongoing clash is an extraordinary degree of creativity.

Immigrants, by their very nature, are entrepreneurial risktakers and drivers of scientific and technological innovation who transfer new skills and knowledge to their adopted states and raise the standards in them. In Israel's early decades, many immigrants were also the hardy survivors of the Holocaust or of long-standing persecution in the Arab world. Others were pioneers, engaged in settling the land and in agriculture in highly adverse conditions, and still others came to Israel despite the opportunity to live a more secure and financially rewarding life elsewhere.[101] All were forced to be creative and seek innovative solutions in order to succeed.

Necessity may be the mother of invention but so too is adversity, whether at the individual or national level. Faced with severe threats to its security and even existence, especially in the early decades when Arab hostility was at its height, resources scarce, and institutional capabilities limited, Israel was forced to find creative ways to survive and build a new society nearly from scratch.[102] Moreover, the pace of change in Israel's environment, both external and internal, has been frenetic. Decision-making in Israel thus came to be all about immediate responses to military threats and staying alive to fight another day, while stitching together the resources needed through a variety of impromptu measures.

The result has been a marked capability to embrace and cope with change, an unusual propensity for improvisation, and a national decision-making style geared toward flexible responses to rapidly changing circumstances, rather than deliberate forethought and systematic planning. To a large extent, this was an unavoidable outgrowth of Israel's harsh external environment, but what began as a necessary evil has remained a primary characteristic of Israeli decision-making to this day, in both governmental and private sectors, indeed, it has become a national hallmark, sphere of excellence, and virtual faith.[103]

Israel's difficult and frenetic environment, further reinforced by the shared experience of military service, has produced a focused and pragmatic problem-solving orientation, and imbued its people with the ability to function comparatively effectively under conditions of high pressure and stress. These are particularly valuable attributes in a field such as the cyber realm, in which the pace of technological development is extraordinary. As Gabi Siboni has noted "the United States has greater capabilities than Israel in cyber space, but we are small and very tense. It is like the difference between a speedboat and an aircraft carrier. We go very fast."[104]

Another view attributes Israel's high-tech success to the freewheeling and unregimented nature of Israeli childhoods, along with parents' greater tolerance for risk in child rearing. The unusual freedom Israeli children have to experiment and take chances, in the absence of strict social norms of behavior, is embedded in Israel's culture and institutions.[105]

Furthermore, and for reasons again having to do with the nature of its early years, Israel's national culture has always been non-hierarchal and informal to the extreme. The era of the kibbutz (agricultural commune) and socialism may be long gone, but the egalitarian norms established by the founders remain, and Israel has been aptly called a "relentlessly informal nation," or in more academic terms, a "small power distance society," in which formal hierarchy is paid little heed.[106] These national attributes are characteristic of the business culture typically found in R&D and high tech firms worldwide. Their informal and flat hierarchical structures and need for a free and open exchange of ideas, in pursuit of collaborative objectives, are highly reminiscent of kibbutz culture.

Israeli culture further combines a well-developed group orientation and willingness to act in pursuit of collective group goals with a strong sense of individualism.[107] Related to this, and again partly a function of Israel's strategic circumstances, is a greater willingness on the part of the governmental, private, and defense sectors to work together in far closer collaboration than is common in most other democracies.

<u>IDF Culture and Cyber Innovation</u>—in addition to the basic national characteristics described earlier, generations of IDF soldiers have been inculcated with the IDF's organizational culture, which emphasizes hard work, tenacity, dedication, professionalism, and team work, all under extremely challenging deadlines and circumstances.[108] Like other militaries, the IDF is highly mission oriented, but it differs in the emphasis it places on creativity and improvisation, no less than the formal authority stemming from rank or education.

IDF officers and even fresh recruits are encouraged to think creatively, out of the box, and to improvise in order to best achieve an objective, rather than mindlessly follow orders dictated from above, even when this means breaking rules. At least by the standards of military organizations, the IDF is informal, flexible, and non-hierarchical, with an unusual tolerance for dispute and risk-taking.[109] Indeed, IDF lore celebrates officers who take the initiative, improvise, and solve problems rather than going by the book. Officers who fail to improvise are commonly perceived as lacking initiative, self-confidence, and resolve. This unorthodox military culture is further reinforced by the ubiquitous presence of reservists, who have little regard for rank, throughout IDF units and echelons.[110]

IDF officers, at all levels, are expected to express their views forcefully when they disagree with superiors and to continue to do so until a final decision is made. Junior officers are expected to press their case with their superiors and senior officers with the chief of staff, who is in turn expected to press his case with the prime minister.[111] Little illustrates this better than the following scene, which took place in the White House just minutes before the signing of the Oslo Accord in 1993, as described by one of the Americans present:

> The commander of the IDF Central Command, Ilan Biran, entered President Clinton's office and launched into a stormy debate with Prime Minister Rabin. The commander argued that Rabin must not agree to the new deployment lines . . . Rabin embarked on a long, vociferous debate with the general, while the rest of us, including the president, had to wait . . . We were astonished. In our wildest dreams we could not have imagined that such a situation would occur between a premier and one of his generals. And all this at the White House, in front of the president.[112]

Unit 8200's organizational culture is reported to resemble that of a startup, and challenges to authority are encouraged. "We teach them how to work out of the box," according to a former senior officer.[113] Unit 81 is even more extreme. Most soldiers in the unit do not wear uniforms, the base does not look like a military installation, and veterans refer to a sense of freedom, "balagan" (chaos), and an emphasis on knowledge and charisma, as opposed to rank.[114]

Unit 81 is essentially a sub-contractor for the intelligence community. The assignments given to it typically appear impossible at the outset, but some veterans view them as the source of their success in later civilian life.[115] According to one veteran:

> What we essentially do each time is establish a startup . . . We solve a problem with a limited budget and at all hours of the day . . . with the understanding that people's lives are at stake. You have one opportunity and can't fail . . . The (unit's) motto[§§] may sound like a slogan, but it is true—if you truly understand a problem and the limitations, you can solve anything. Later on in the business world, it changes the way you

[§§] "Turning the impossible into the possible."

> look at matters . . . and the importance of providing a precise solution to the requirements and testing to make sure that it works consistently.[116]

Yigal Unna, a former head of the INCD, stresses that the IDF gives soldiers "a license not only to do interesting things for the state in the cyber arena, but also trains them for one thing: not to be afraid, to try, to take risks, to experience failure."[117] The officer in charge of the IDF's cyber defenders course states that soldiers are taught "to think differently, to search for things that are less visible, in the places where one does not usually look. They have to search for things on the Internet that people have tried really hard to hide. If we just search like everyone else, we won't find them."[118] A former commander of Unit 81 described its recruitment and training programs in similar terms:

> When the soldiers are recruited the emphasis is not necessarily on their knowledge of computers or electronics. We are looking for people who can think outside the box, but who can also collaborate with others who have similar traits . . . we don't ask candidates to write code, instead we give them a complex problem to solve in order to examine how they come up with a solution.[119]

The shared military experience further engenders a strong sense of group cohesion and a desire to continue working together in civilian life, one of the reasons so many high tech firms in Israel have been established and staffed by veterans of units such as 8200 and 81. After leaving Unit 81, according to one veteran, "you realize that up to that time you did the wildest things in your life together with the most talented people and you want to extend that into civilian life." A former commander of Unit 81 observes that soldiers still doing their compulsory service "see veterans doing well and don't want to become salaried developers, but to blaze new paths, because that was the drive that was instilled in them."[120]

Units 8200 and 81 may be unusual, but they do reflect something unique about the IDF in general. In any event, the willingness to challenge authority and conceptual orthodoxies, internalized prior to and during military service, has a direct impact on Israel's civil high tech world.[121]

Social Networks

Social networks, domestic and international, are important determinants of scientific and technological achievement. Social networks can provide information about a variety of critical areas more efficiently than markets or governments, including scientific and technological developments, job opportunities, and ties to skilled applicants. They can further match entrepreneurs and investors who might otherwise not find one another and help find business opportunities for highly specialized high tech products. In so doing, social networks drastically reduce the costs and risks of innovation.[122]

Social networks are especially important sources of information about geographically removed opportunities, of particular importance for Israel given its distance from international markets. To this day, but especially in the 1980s when its high tech sector was still in its infancy, Israel has built networks linking Jewish businesspeople and financiers in the United States with high tech startups in Israel. A further international network was formed by immigrants from scientifically and technologically advanced countries, whether from the West or especially Russia during the 1970s and 1990s.[123] Diasporas are particularly efficient networks for connecting people and exchanging information and knowledge required for innovation.[124]

Clusters are a particular form of social network. Clusters produce valuable exchanges of information that help firms become more competitive, provide skilled labor and economies of scale, attract FDI, and have a variety of valuable assets, including technology, skills, and information about markets and customer needs. Clusters are at their best when they allow scientific and technological personnel to move easily between firms or create spinoffs and startups of their own. Universities and research institutes are often essential players in clusters.[125]

Israel remains a small country, with a population that is now similar in size to that of New York City, Switzerland, or Austria. The old Israeli truism, whereby "everyone knows everyone" whether from high school, military service, or university, is no longer quite true, but is not off by that much. The technological expert whose advice one seeks is usually no more than a phone call away[126] and national decision-makers are typically at a distance of no more than two degrees of separation. There may be a number of identifiable high tech hubs, but the physical distances are small and in practice all of Israel, between Haifa and Beersheba, really constitutes one long cluster—a "Silicon Wadi." The result is an easily accessible wealth of expertise and ease of communication that greatly facilitates R&D processes in Israel.

Virtually everyone in Israel's high tech sector has served in the IDF, many in the technological or combat units. When combined with the nation's small population and geographic size, the result is a comparatively close relationship between the government, IDF, academia, and high tech sector. Reservists, in particular, serve as a conduit and bridge between them.[127] Reservists gain firsthand exposure to the IDF's operational needs and shortfalls and thus the ability to propose ideas for new or improved capabilities. Many reservists also have close ties to decision-makers at senior business and national levels, thereby greatly reducing the time between the identification of a need and development of the necessary response. Reservists further form an important network for exchanges of professional and social information and for recruitment purposes, matching the needs and skills of employers and employees.[128]

The opportunity to maintain connections with peers, former commanders, and reservists helps create a highly developed networking system for business, employment opportunities, and social purposes. A single individual's military service can yield hundreds of ties on various levels and with people of diverse localities and socioeconomic backgrounds.[129] Some headhunting firms specialize exclusively in veterans of Unit 8200 and/or other technological units in the military. Unit 8200 itself has an alumni association of 15,000 veterans, which hosts a variety of networking events designed to help members find appropriate employment in the R&D sector and to promote business opportunities. Unit 81's alumni association has 5,400 members and conducts various programs designed to develop entrepreneurial skills, help those interested in entering the entrepreneurial world, and promote entrepreneurial endeavors. Unit 9900 has a similar network.[130]

Veterans of Unit 81, for example, tend to recruit their former subordinates and teammates when launching startups. They in turn leave these firms to found startups of their own, recruiting people discharged more recently from military service, and creating a repetitive cycle. Indeed, it is not rare for firms to be founded by teams comprised entirely of Unit 81 veterans. The networking benefits do not end with the founding of firms or recruitment and mentoring of newly discharged comrades. Younger veterans commonly raise capital from their predecessors in an informal angel circuit.[131] Veterans of Unit 8200 and other high tech units act in similar ways.

Clouds, Possible Misdeeds, and New Opportunities

Israel's many achievements in the high tech and cyber areas notwithstanding, it faces a number of difficult challenges in the coming years. With a small population base to draw on, a shortage of computer scientists, engineers, and other highly skilled professionals has become a major obstacle to further expansion of Israel's high tech sector, especially in the cyber realm. By the mid-2020s, it has been estimated that Israel will face a shortage of approximately 10,000 engineers and programmers in a market that currently employs 140,000.[132] In 2019 Israel already suffered from an overall shortage of 18,500 tech workers (engineers, programmers, and others)[133] and in the cyber realm alone a shortage of some 20% of the personnel needed.[134] Competition with global giants such as

Facebook, Google, and Microsoft, which offer especially lucrative employment packages, makes the competition for highly skilled personnel particularly fierce.

At present, the high tech sector is still benefiting from the major strides made by Israel's educational system in earlier decades. In the coming years, however, the ongoing deterioration of the educational system that has taken place in more recent decades, at all levels, will presumably have a growing impact.[135] In 2020, the head of the National Council for R&D warned that Israel may lose its leading place in innovation and high tech by 2030 if it does not greatly increase funding for academic R&D. On paper, Israel invests a great deal in R&D, per capita, especially compared to Europe, but 85% of Israeli R&D, he argues, is actually conducted in the R&D centers established by the multinational corporations, while government funding for academic R&D is now among the lowest in Europe. Whereas the budget of the Israel Innovation Authority equaled 1% of the state budget in the early 2000s, it constituted less than 0.5% in 2020, admittedly of a much larger state budget; the EU, United States, and South Korea devote between 0.6–1% of their GDPs to innovation, Israel now devotes only 0.15%. The multinationals follow the human resources, and if they conclude that there are better opportunities in other countries, R&D investment in Israel may dry up rapidly.[136]

Contrary to the public image, emigration from Israel is low, especially for an immigrant society, indeed, lower than the OECD average,[137] and has decreased steadily over time. Even among highly qualified native-born Israelis, emigration is just above the OECD average.[138] Conversely, the quality of the human capital lost is high and Israel is suffering from an ongoing brain drain. In 2015, 5.6% of all Israelis who had received undergraduate or graduate degrees between 1980 and 2009, in any field, had lived abroad for three or more years. The problem was particularly acute among PhDs, of whom 11% had lived abroad for three or more years, and especially so among those with PhDs in mathematics (25%), computer science, aeronautical engineering, biology, chemistry, physics, and genetics (each between 16–18%).[139] Obviously, the three-year cutoff point is arbitrary, and many will ultimately return, but it provides a clear indication of a worrisome trend for Israel.[140]

Given the growing shortage of high tech personnel, Israeli firms are increasingly setting up R&D facilities off shore, with Ukraine, Russia, and India the most popular sites.[141] In 2017, for the first time, the government was forced to approve the hiring of 500 foreign high tech workers. Plans were also announced that year for a 40% increase in the number of computer science graduates over five years, at a cost of some 700 million shekels. In 2018, in an indication that the five-year plan was beginning to pay off, and for the first time in Israel's history, more students registered for engineering than for any other academic discipline. Together with computer science, mathematics, and statistics, they accounted for approximately one-quarter of all university students in Israel. In early 2022, in a further indication of the five-year plan's success, the National Economic Council found that the number of students in the high tech disciplines had actually increased by 50% and forecast that this would continue to grow by an impressive 40% to a whopping 25% by 2030. Favorable demographic trends, especially the rapid growth of Israel's young secular population, the primary source of high-tech personnel, and an increase in the number of high school students studying advanced mathematics, were among the reasons for the more optimistic assessment.[142]

The success in bringing new companies to the ATP in Beersheba in the early years notwithstanding, the city's geographic distance from the center of the country has proven a major stumbling block, and the pace of expansion has slowed. Even relocation of IDF intelligence units has been postponed indefinitely due to internal resistance. The government plans on launching a number of new programs to help promote the ATP, but its stated objective of reaching 10,000 high tech employees in Beersheba by 2025, not including IDF personnel, appears ambitious.[143]

Although the overall number of cyber firms in Israel doubled between 2014 and 2019 and the total value of venture capital raised by the cyber sector, as well as cyber IPOs, have grown steadily, the rate of growth has slowed significantly, and Israel's cyber industry appears to be maturing. Whereas 86 new cyber firms were established in 2015, only 70 were established in 2017, 42 in 2018, and just 12 in 2019. The number of new multinational R&D centers established in Israel has similarly slowed, from 40 in 2015 to 23 in 2019 and just 4 in 2020. This decrease mirrors the slowdown in the high tech sector as a whole, from 1,400 new startups established in 2014 to just 520 in 2020.[144]

One study found two primary reasons for the dramatic decrease in the overall number of high-tech startups in Israel. The first reason, which accounts for most of the change, is typical of the global high-tech arena as a whole. Whereas advertising and social media had been the primary engines behind the establishment of new startups between 2010 and 2014, following the advent of smart phones, changes in technology, market forces, and governmental regulation led to a drop in the number of new startups in these areas. The second, more Israel-centric reason, has to do with the rapid increase in the number of global corporations operating in Israel, approximately 200 during the same 2010–2014 period. These corporations offer highly attractive, low risk compensation packages, thereby reducing the motivation to establish new start-ups.[145]

Israel has, therefore, decided to invest heavily in what it believes will be the next major realm of innovation, AI. In an effort to replicate the successful model that led to Israel's cyber revolution, a special task force was established to propose a national AI strategy, once again with the participation of the government, IDF and defense establishment, academia, and the high tech industry. The task force submitted its recommendations in 2020. Much as its cyber predecessor had done years earlier, the AI task force recommended that Israel seek to become one of the five global leaders in the field within five years. Israel is already thought to be in third place in the number of AI related firms, after the United States and China, and sixth in leading AI scientists.[146] The task force further recommended that a special body, akin to the INCD, be established to lead and coordinate all national efforts in the AI area and that a budget of some 10 billion shekels be allocated over five years.[147]

In the field of quantum computing, in contrast with AI, a special committee recommended in 2020 that Israel only seek to become a "threshold state," given the magnitude of the investment required to become a world leader. Nevertheless, the government allocated 1.25 billion shekels for a five-year National Quantum Initiative, largely to develop the necessary human resources, as well as 200 million shekels to build a foreign-made quantum computer in Israel, which was to constitute the basis for the future construction of an Israeli-made one. The defense establishment, which attributes strategic importance to quantum computing, is deeply concerned that states that do not have their own independent capabilities in this field may not be able to purchase them abroad. It is thus heavily involved in the initiative.[148]

Software sold by Israeli companies is currently believed to be in use in roughly 130 states.[149] Over and above their commercial value, a significant consideration for all states, these sales have also become an important part of Israel's efforts to build relations with foreign partners, including those with which it did not previously enjoy diplomatic ties. This has been particularly true of a number of Arab states, which have overcome their historic hostility and established commercial and even formal diplomatic ties with Israel, at least partially in order to gain access to its high tech and cyber technology.[150] For diplomatically challenged Israel, this new form of "cyber diplomacy" has become an important instrument of foreign policy. It paid off particularly handsomely in the decision of Bahrain and especially the UAE to sign peace agreements and establish diplomatic relations in 2020, which have been greatly expanded in a variety of areas ever since (see Chapter 9).

In the past, Israel restricted cyber exports to defensive software, banning offensive sales, a highly ambiguous distinction in the cyber realm.[151] In effect, it left it up to the exporters to make the distinction, a clearly unsatisfactory situation. Israel subsequently began also approving sales of offensive software, leading to more criticism from various human rights groups, multinational corporations, and others who argued that the country was already too lenient and demanded that Israel adopt tighter controls. Israeli firms, conversely, contended that governmental export controls were stricter than those in most other countries and placed them at a competitive disadvantage.[152] One government official put the dilemma this way: were Israel to increase oversight too much, these firms would simply move to Cyprus or Macedonia, and it would lose both the firms themselves and the ability to supervise their exports.[153]

A key component of Israel's export laws revolves around the Wassenaar Arrangement, a nonbinding international arrangement between 42 nations that regulates the export of dual-use technologies, including cyber. Inclusion of a technology on the Wassenaar Arrangement "control lists" does not constitute a ban or prohibition on its export but is a commitment by the participating state to require that sales be governed by its national export control policy and licensing requirements. Most of the member states, including those from the EU, have added certain surveillance and intrusion software to the regulated items. The US position, in contrast, remains to be resolved, due to ongoing concerns regarding the differentiation between offensive and defensive uses. Israel is not a party to the Wassenaar Arrangement but adheres to its guidelines and has passed laws adopting them.[154]

This ambiguity between offensive and defensive uses has led to some troubling outcomes in Israel's case. Cyber tools from Israel have been exported to authoritarian regimes, which have reportedly used them to target journalists, dissidents, human rights activists, and others. Two firms, in particular, have been the focus of attention, the NSO Group and Verint, both of which maintain that they sell spyware only to governmental clients and solely for purposes of counter terrorism and crime prevention. Their clients, however, include noted human rights violators, such as Saudi Arabia, Azerbaijan, Mexico, the United Arab Emirates, Qatar, Bahrain, South Sudan, Indonesia, and more.[155]

In 2021–2022 NSO became the focus of a global scandal, with ramifications for Israel's international standing and use of cyber as an instrument of diplomacy. Seventeen media organizations, including the *Washington Post*, the *Guardian*, *Le Monde*, and *Haaretz* joined together with Amnesty International and a French media nonprofit to form the "Pegasus Project," an international investigative consortium named after NSO's premier product. In a series of detailed reports, blasted over a number of consecutive days on the consortium members' front pages, as well as by other leading news media around the world, such as the *New York Times*, the Pegasus Project presented a dramatic picture of alleged abuses by NSO and other Israeli cyber firms. The fact that NSO, according to the *Washington Post* and others, is just one of a number of major firms in its field further accentuated the unique attention afforded to it.

The consortium found that 189 journalists, dissidents, human rights activists, politicians, and heads of state, in 21 countries had been identified as possible targets of surveillance by at least 12 NSO client-governments, and that this surveillance had actually been carried out in a number of cases. Those targeted came from a list of over 50,000 phone numbers from approximately 50 countries, which may have been noted as subjects of interest and considered for surveillance by NSO clients. The consortium was able to identify more than 1,000 people on the list, including six sitting presidents and premiers (in France, Iraq, South Africa, Pakistan, Egypt, and Morocco), seven former premiers (France, Belgium, Yemen, Lebanon, Algeria, Uganda, and Kazakhstan), and the King of Morocco. Also on the list were the phone numbers of the wife and fiancé of murdered Saudi journalist Khashoggi and a prominent Mexican reporter who was gunned down on the street, indicating the possibility of indirect NSO culpability. The phones of 11 US diplomats serving in Uganda were also hacked using Pegasus.[156]

NSO's close ties to the Israeli government and the allegation that it had provided the government with at least some of the intelligence collected were particularly damaging. In some cases, such as India and Hungary, NSO clients reportedly started using Pegasus just a short time after state visits by Prime Minister Netanyahu and possibly as a direct result. Moreover, the MoD was reportedly involved in selecting, sponsoring, and in some cases even initiating the first contacts with NSO clients, including active encouragement of ties with the UAE, Bahrain, and Saudi Arabia, even after the Khashoggi scandal erupted. The government reportedly also encouraged a number of other Israeli cyber firms[***] to work with Saudi Arabia and other clients.[157]

Following the revelations, a number of states, including the United States, UK, France, Canada and Australia as well as the EU and leading multinational firms, such as Apple and WhatsApp (which is owned by Facebook), expressed concern about Israeli cyber sales to authoritarian regimes. The United States ultimately blacklisted NSO and Candiru, another one of the Israeli firms, from receiving exports from US firms,[158] even though critical US security agencies had actually either bought Pegasus or considered doing so. The CIA, for example, bought Pegasus on behalf of the government of Djibouti, for counterterrorism purposes, despite long-standing concerns regarding human rights abuses. The FBI bought it for possible use in criminal investigations, but claimed that it had merely evaluated the program and decided in the end not to make use of it, while the DEA found Pegasus too expensive. The Secret Service and US Africa command held discussions with NSO regarding Pegasus.[159]

These incidents and others have led a number of human rights groups, as well as Microsoft, Facebook, and others, to call upon Israel to more tightly regulate its cyber exports and in NSO's case ban them completely. Even before the Pegasus Project scandal erupted, Amnesty International, Facebook, and 50 other entities

[***] Verint, Candiru, Quadream, and Cellebrite.

had filed suit against NSO and the Defense Ministry, arguing that Israel had failed to meet its duty to halt exports when evidence of harm existed.[160]

Approximately half a year after the international scandal, a domestic uproar erupted in Israel over reports that the police had made unauthorized use of Pegasus against some two dozen prominent Israelis. The police were accused of either bypassing judicial oversight altogether or of deploying spyware from NSO and other firms in a manner that exceeded the parameters set by the courts. A government inquiry subsequently found that the police had only actually exceeded the court orders in one, possibly two, cases and that no use had been made of the information collected.[161] The brief uproar, nevertheless, brought the dangers of unbridled use of spyware of this sort home to the Israeli public, turning it from a foreign policy issue to one with domestic ramifications as well.

Under the combined weight of the scandals, domestic and international, NSO, and a number of other companies encountered growing financial difficulties, A number closed entirely and NSO downsized significantly. Its long-time founder and CEO was forced to resign.[162]

Even before the NSO scandal, the government had begun responding to the conflicting commercial, diplomatic, and legal needs, streamlining the export approval process but also adopting measures to ensure greater oversight.[163] One key component was shortening the length of time it takes Israeli firms to go through the rigorous export license process from the commercially untenable year or longer to a few months.[164] Conversely, a new division was to be established in the Ministry of Economy and Industry to strengthen oversight of exports of cyber technologies with civilian applications and to complement the already existing mechanism in the MoD, which is responsible for oversight of security related exports. The MoD cut the number of states eligible to buy offensive cyber exports from 102 to 37, mainly the United States, Canada, the EU, Japan, India, Australia, and New Zealand. Saudi Arabia, the UAE, Morocco, and Mexico were no longer included. Sales were to be limited to governmental customers and to be used solely for purposes of the investigation and prevention of terrorism and severe crimes. The definition of terrorism and severe crimes was explicated, to prevent misuse, and legal sanctions were established for violations of the export controls, which were based on the "Wassenaar agreement." The new export controls also specifically defined those uses which would be proscribed, including those that might cause damage to people on the basis of religion, gender, race, national origin, and political and sexual orientation. The IDF began exempting reservists who work for firms engaged in offensive cyber tools from reserve duty, thereby reducing the danger of them being exposed to highly sophisticated new capabilities.[165]

Israel is not the only country trying to figure out how to deal with the benefits and dangers posed by cyber exports. The United States, UK, and other nations play a big role in these markets and are struggling with similar issues. For instance, even though the United States is a signatory to the Wassenaar Arrangement, it has been slow to modify its export laws to become compliant, whereas Israel did so in 2013.[166] In addition, US, British, German, Italian, Austrian, and Greek firms have engaged in behaviors similar to NSO and Candiru, including sales to some of the same problematic states as Saudi Arabia, the UAE, China, and Belarus. In some cases, former members of the US national security establishment were employed by these firms and, much as in the case of the sales by Israeli firms, allegedly targeted human rights activists, journalists, and political rivals.[167]